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The economics of innovation: from the classical legacies to the economics of complexity

Cristiano Antonelli [a]

[a] Dipartimento di Economia, Università di Torino, BRICK (Bureau of Research in Complexity, Knowledge, Innovation), Collegio Carlo Alberto, Italy

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The economics of innovation: from the classical legacies to the economics of complexity

Cristiano Antonelli*

Dipartimento di Economia, Università di Torino, BRICK (Bureau of Research in Complexity, Knowledge, Innovation), Collegio Carlo Alberto, Italy

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During the last 40 years, economics of innovation has emerged as a distinct area of enquiry at the crossing of the economics of growth, industrial organization, regional economics and the theory of the firm, becoming a well-identified area of competence in economics specializing not only in the analysis of the effects of the introduction of new technologies, but also, and mainly in understanding technological change as an endogenous process. As the result of the interpretation, elaboration and evolution of different fields of analysis in economic theory, innovation is viewed as a complex, path-dependent process characterized by the interdependence and interaction of a variety of heterogeneous agents, able to learn and react creatively with subjective and procedural rationality.

Keywords: knowledge; innovation; technological change; diffusion; path dependence; complexity

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1. Introduction

During the last 40 years, economics of innovation has emerged as a distinct area of enquiry at the crossing of the economics of growth, industrial organization, regional economics and the theory of the firm, and has become a well-identified area of competence in economics specializing not only in the analysis of the effects of the introduction of new technologies, but also, and mainly, in understanding technological change as an endogenous process.

As a result of the interpretation, elaboration and evolution of different fields of analysis in economic theory, innovation is viewed as a complex, path-dependent process characterized by the interdependence and interaction of a variety of heterogeneous agents, able to learn and react creatively with subjective and procedural rationality.

After the discovery of the residual, the traditional assumptions about the exogeneity of technological change proved to be quite embarrassing and pushed economics to provide and elaborate an endogenous explanation: too large is the share of unexplained growth. Neoclassical economics provided an elaborated and sophisticated framework to understand the conditions for static efficiency. In such a context, growth and development are the

*Email: cristiano.antonelli@unito.it

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consequences of exogenous changes in the shape of utility functions, in the characteristics of the technology, and in the demographic conditions as well as the supply of natural resources. This theory does not address the actual causes of growth and change. It is limited to analyzing the complementary conditions in terms of rates of growth in the supply of labor and savings that make it possible to take advantage of the effects of 'technology push' falling from heaven like manna and for exogenous growth to take place.

The effort to provide an endogenous explanation of technological change has been nurtured by the sequential and yet overlapping articulations and reinterpretations of different approaches that have been progressively built in a process of reconsideration and reappraisal of the dynamic legacies that had stressed the role of endogenous dynamics, but had been left aside by mainstream theorizing.

Four wide-ranging heuristic frameworks can be identified: the classical legacies of Adam Smith and Karl Marx, the Schumpeterian legacy, the Arrovian legacy, and the evolutionary approaches leading to complexity.

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These four approaches have a clear focus. After the discovery of the residual, the economics of technical progress attempted to extract from the classical legacies an endogenous explanation of growth. The economics of technical change was articulated around the induced approach to technological change derived from Marx, and the demand-pull approach implemented in a post-Keynesian context upon the line of enquiry based upon Adam Smith as interpreted by Allyn Young. Both have been especially useful in understanding the contribution of technological change to economic growth, mainly at the aggregate level with a strong emphasis upon structural change. The microeconomics of innovation, however, was completely missing. The Schumpeterian legacy has provided the basis of enquiry into the relationships between innovation and competition in the marketplace with important implications for the theory of the firm and the theory of the markets. The Schumpeterian approach has focused on the role of innovation as a competitive tool, and on both the corporation and entrepreneurship as the driving factors. The Arrovian legacy has made it possible to explore the economics of knowledge with its powerful implications for the theory of organization and regional economics. Finally, biological grafting based on the Marshallian legacy and the evolutionary approaches, lately reinvigorated by complex thinking, has paved the way to understanding the path-dependent dynamics and systemic interdependencies that characterize technological and structural change.

The approaches have evolved in parallel in the second half of the twentieth century with a process of specialization and consolidation of their respective areas of expertise. Yet an increasing number of lateral and horizontal contributions have been made, feeding a process of increasing convergence and integration. As a result, a rich web of overlapping stratifications has gradually been growing.

In order to highlight the origins and the evolution of the economics of innovation, a matrix of analytical tools can be elaborated. The basic line of understanding is found along the diagonal, where each field matches its own basic approach. Much interest, however, is found in cells around the diagonal, where an increasing number of cross-contributions can be identified. The result of the process is an increasing complementarity and compatibility among the four approaches into the new frame provided by the economics of complexity. Table 1 synthesizes the matrix of notions and concepts elaborated in the economics of innovation. It shows how the different analytical trails have contributed to the evolution of the field.

The classical, Schumpeterian, Arrovian and evolutionary approaches share a basic departure from standard economics: the attribution to economic agents with the capability to change their production and utility functions. A few common threads also emerge across

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Table 1. The innovation matrix.

	Innovation and growth	Innovation and competition	Innovation and knowledge	Innovation within evolving systems
The classical legacies	Division of labor Demand-pull Inducement		Learning Collective knowledge	Industrial specialization Variety Structural change
The Schumpeterian legacies	Creative destruction	Creative reaction The Schumpeterian hypothesis Entrepreneurship Monopolistic competition Structure-conduct-performance Dominant design Network externalities	Gales of innovation R&D Technology push Technological opportunities	Dynamic efficiency Sectoral patterns Technological regimes Creative adoption General-purpose technologies
The Arrovian legacy	New growth theory	Knowledge as a production factor Knowledge quasi-rents Spillover	Knowledge as an economic good Knowledge spillover Industrial districts Knowledge asymmetries Knowledge governance	Technological systems Systemic interactions
Evolution and complexity	Learning as the engine of growth Technological trajectories	Life cycle Epidemic diffusion Replicator dynamics	Localized technological knowledge Distributed knowledge Innovation networks Knowledge as an input and an output Competence	Localized technological change Past dependence Positive feedbacks Path dependence Generative relationships

them: first, the heterogeneity of learning and interaction conditions; second, the effects of historic time, both at the system level and at the agent level; third, the conduct of each agent is shaped by the effects of the past and yet they are credited with the capability to alter the trajectories of their activities by means of the generation of new technological knowledge and the introduction of technological innovations. The dynamics of feedback is finally appreciated in the different contexts: the introduction of innovations changes the structure of the system and this in turn affects the conduct of agents, including the introduction of other innovations.

The shared understanding of the working of the economic system as a complex dynamic process is possible as soon as the systemic properties that belong to economics as a science are extended so as to include the possibility for agents and subsystems to internally generate new technological knowledge and hence new production technologies and new preferences. What is more, it is not difficult to do this, for most of standard microeconomics can be retained and properly implemented at the agent level: heterogeneous agents do attempt to optimize within the strong limitations of their subjective conditions. The aggregate outcome of their action, however, is far from a general equilibrium steady state, and rather a process of continual change typical of complex systems dynamics.

This view can be considered the result of the integration of the four approaches and of the four fields of investigation into one broader analytical platform provided by complexity economics. With respect to Table 1 we see that, especially at the bottom right of the diagonal, the borders of the cells themselves are more and more blurred as systematic overlapping across fields of investigation and traditions of analysis take place. Economics of innovation emerges as a distinctive field of investigation with a broad array of complementary concepts articulated within convergent and consistent fields of specialization.

The aim of this work here is to provide an interpretative framework able to identify the main contributions of the economics of innovation as a pathway to the emerging economics of complexity. To do this, we shall try and track the emergence of the view that innovation is a path-dependent, collective process that takes place in a localized context if, when and where a sufficient number of creative reactions are made in a coherent, complementary and consistent way. As such, innovation is one of the key emergent properties of an economic system viewed as a dynamic complex system (Antonelli 2008a).

2. Innovation and growth – the reappraisal of the classical legacies

2.1. The discovery of the residual

The discovery of the residual coincides with the birth of economics of innovation. In neoclassical economics, technological change is exogenous. Occasionally, technological shocks perturbate the equilibrium conditions of the system: firms are not supposed to be able to intentionally change their technologies. The empirical investigations of Solow (1957) and Abramovitz (1956) show that over 50% of the growth of output in the US economy between the end of the nineteenth century and the first part of the twentieth century cannot be reconciled with the growth of inputs. Technological change should be credited for its astonishing contribution to economic growth. Equilibrium economics is able to explain only a fraction of the economics system. It is this evidence that presents a challenge.

The basic puzzle remains the problematic core of this area of specialization. How innovations come to the marketplace, how novelty takes place in our understanding of the economic and technological interplay, how and why total factor productivity grows, how firms and economic agents in general generate and react to the introduction of novelty, are the key questions. The birth of economics of innovation as a specific area of enquiry and

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investigation in the broader context of the increasing specialization of economics can be considered the ultimate result of the analysis of growth of output and labor productivity, *ceteris paribus* input levels, when and if increasing returns do not take place. As will be shown below, it was the writings of Adam Smith and Karl Marx that provided the key points of departure to the economics of innovation and complexity.

2.2. Adam Smith and the demand-pull hypothesis

In the effort to grasp the determinants of the early stages of the first industrial revolution, Smith (1776) contributed the basic elements of the demand-pull approach. The division of labor is determined by the extent of the market. All increases in the extent of the market can lead to an increase in the division of labor and hence in specialization. Specialization in turn is the base for dedicated learning and the eventual introduction of innovations. Innovations increase the efficiency of labor and hence the extent of the market. As Rosenberg (1965) notes, Smith laid down the foundations for the analysis of technological change as an endogenous and self-feeding process. Young (1928) and Kaldor (1972) have developed this approach further, and shown the irrelevance of equilibrium economics when the dynamics of technological change is understood.

Smith provided the most impressive and clear account of the essential role of technological knowledge and technological change as endogenous factors in explaining the dynamic character of the economic process. The first four books of *An inquiry into the nature and causes of the wealth of nations*, the founding stone of economics, are fully devoted to exploring the economic process and its determinants. The very first lines of *An inquiry into the nature and causes of the wealth of nations* are devoted to the relationship between productivity and the division of labor. According to Smith, the growth in productivity is a consequence of the division of labor: 'The greatest improvement in the productive powers of labor, and the greater part of the skills, dexterity, and judgment with which it is anywhere directed, or applied, seem to have been the effects of the division of labor' (Smith 1776, 13).

The division of labor has a clear causal role in Smith's view of the origins of the accumulation of competence and knowledge. More specifically, Smith elaborated a sequence according to which the division of labor is the cause of an increase in competence. The generation of new knowledge builds on the increase in competence, and technological innovations are the result of this process. Smith fully articulated a bottom-up theory of technological knowledge. Learning by doing and learning by using are at the origin of inventions that eventually make possible the introduction of new and improved machineries.

According to Smith, the professional competence of workers is acquired and implemented by means of learning processes that, ultimately, because of the division of labor, are the cause of the skills of workers. However, learning by doing and by using processes, internal to each firm, are not the sole factors in the accumulation of new knowledge. An important role is also played by the producers of machines and by scientists. The division of labor in conclusion, enters the workings of science and becomes a powerful factor in the organization and efficiency of scientific progress.

A reading of Smith confirms the key role of the economics of knowledge in the understanding of the economic process shaped by continual development based on the introduction of new technologies. Indeed, one finds in Smith the early foundations of the economic understanding of the mechanisms at work in the generation of technological knowledge. Smith in fact provides a comprehensive analysis where technological knowledge is regarded as the eventual result of at least three processes: (a) a bottom-up process by means of learning by doing and learning by using; (b) the specialized activity of

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'philosophers' in a top-down process; and (c) the interactions with suppliers of machinery and intermediary inputs.

Building on these bases, the dynamic engine of Smith is in place. The division of labor is the consequence of the extent of the market and is the cause of the increase of technological knowledge, hence of inventions and eventually technological innovations. Technological innovations in turn lead to an increase in productivity. The increase in productivity leads to an increase in the demand and hence of the extent of the market. Thus the analysis of Smith comes full circle.

Alfred Marshall followed the line of enquiry elaborated by Smith and acknowledged the dual relationship between the division of labor and the introduction of new technologies. Technological change and specialization are two sides of the same process. Allyn Young, following Marshall, probably contributed to focusing attention on the key role of endogenous dynamics in the work of Smith. According to Young, the interaction between technological and structural change is fed by the dynamics of division of labor. Specialization, accumulation of competence, introduction of new technologies and further increase in the extent of the market are the steps of a progressive and cumulative change. In so doing, Young captures the critical role of technological change, as both the product and the cause of increasing functional differentiation and complementarity within the economic system, in economic growth. He lays down the first elements of a system dynamic approach to understanding economic growth. Economic systems in fact are viewed as complex and dynamic adaptive organizations composed by autonomous and yet interrelated and interdependent units that change over time.

Nicholas Kaldor digs even deeper and fully recognizes the essential contribution of Smith to building a dynamic theory of the economic process where technological change and technological knowledge pulled by the interplay between the beneficial effects of the division of labor and the extent of the market take center stage. Cumulative technological change takes place, in out-of-equilibrium conditions, in an economic system where and when firms are viewed not as passive users of given technologies only able to select the techniques more appropriate to a given set of relative prices, but as agents able to change and generate their own technologies.

Building on Smith, Schmookler's (1954) empirical support to the hypothesis that demand growth pulls the increase of technological knowledge, hence of inventions and eventually technological innovations. Mowery and Rosenberg (1979) provide an outstanding account of the pervasive role of the demand-pull hypothesis within the post-Keynesian approach.

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2.3. Karl Marx: induced technological change and the early economics of knowledge

The economics of innovation owes to Karl Marx at least two lines of enquiry: the analysis of technological change as the intentional result of a meta-substitution process induced by the changing cost of labor, and the analysis of the generation of technological knowledge as an economic process.

2.4. The inducement mechanism

Building upon a superb command of the historical processes that characterized the first industrial revolution, Marx contributed the first elements of a theory of endogenous technological change as the result of the intentional process of augmented labor substitution. When wages increase, capitalists are induced to introduce new capital-intensive

technologies that help to reduce the pressure of unions and increase the total efficiency of labor (Marx 1867/1976).

Technological change in fact is the basic tool by which capitalists increase profits together with the extraction of surplus value from the production process. Marx stressed the dual role of technological change. On the one hand, technological change makes it possible to reduce the price of goods in the marketplace. On the other, technological change makes it possible to increase the extraction of surplus value (Rosenberg 1976). The competitive process among capitalists feeds the former. The exploitation of labor by capitalists as a class is the result of the latter.

Rosenberg (1976) provides an admirable account of the dynamics of the innovative process in the Marxian approach. The attempt to contrast the decline in the profitability of each firm, stemming from the increase in wages, is the direct incentive for the innovative action of each capitalist that leads to the generation and introduction of technological innovations embodied in new machines. The competitive pressure in the markets for products, among capitalists, adds momentum to the dynamics of the process. Marx paid a much stronger attention to process innovations rather than to product innovations: hence at the system level, the innovative process has the aggregate effect of substituting capital for labor. The decline in the number of workers is relative, but not absolute. After the absorption of temporary unemployment wages increase, again, and hence capitalists are, again, induced to introduce new process technologies that reduce the labor intensity of the production process. As Rosenberg (1969) argues, Marx sets forth the inducement hypothesis that characterizes much of the economics of innovation.

Specifically, a distinction has to be made between models of induced technological change, which focus on the changes in factor prices, and models of induced technological change, which stress the static conditions of factor markets. In the first approach, following John Hicks and Karl Marx, firms are induced to change their technology when the price of a production factor increases (Hicks 1932). According to Hicks, the change in factor prices acts as a powerful inducement mechanism that explains both the rate and the direction of the introduction of new technologies. The change in factor prices in fact induces firms to introduce new technologies to reduce the cost of the factor that has become more expensive. The introduction of new technologies complements the standard substitution process, i.e. the technical change represented by the selection of new techniques, defined in terms of factor intensities, on the existing isoquants. In this case, as the Fellner (1961) technological change is considered an augmented form of substitution: technological change complements technical change.

As the debate summarized by Binswanger and Ruttan (1978) shows, this approach to induced technological change differs from the static Kennedy–von Weizsacker–Samuelson approach, according to which firms introduce new technologies in order to save on the production factors that are relatively more expensive. Here the levels of factor price matter instead of the rates of change. This approach has revealed a major limitation of the former. From simple algebraic calculation, it is in fact clear that firms have an incentive to introduce labor-intensive technologies, in labor-abundant and capital-scarce regions and countries, even after an increase in wages. The Kennedy–von Weizsacker–Samuelson approach, however, is severely limited from the dynamic viewpoint. It is no longer clear when and why firms should innovate. Consistently, only the direction of technological change can be induced, rather than the rate. Both approaches, as is well-known, have often been criticized using Salter's argument, according to which firms should be equally eager to introduce any kind of technological change, either labor- or capital-intensive, provided it enables reduced production costs and increased efficiency.

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It is interesting to note that the analysis of the role of relative factor endowments in explaining the direction of technological change has recently been revived, by Acemoglu (1998, 2002), to explain the bias toward new information and communication technologies in terms of skill intensity.

The re-appraisal of this tradition of analysis, known in the literature as the economics of technical progress, and its integration into the economics of innovation is necessary in order to grasp the evolutionary outcome of the interaction between the introduction of new technologies and the changes brought about in the economic system. Innovation is but a part of a broader process of heterogeneous interactions between the effects and the determinants of technological and structural changes, which takes place in a disequilibrium context (Day 1983).

The classical legacies show how the understanding of technological change cannot be separated from the analysis of structural change. Technological change and structural change are two intertwined and inseparable parts of a single process of change and growth that takes place in historic time. Along these lines, David (1975) shows that only when historic time is taken into account, it is possible to understand that change is both the cause and the consequence of the introduction of innovations. The long-term analysis of economic growth shows the persistency of factor intensity presumably explained by elastic barriers based on local irreversibilities and switching costs that prevent firms adjusting to the changing levels of relative input costs. When significant changes in the relative costs of production factors and in the overall levels of demand take place, firms react with the introduction of new technologies. The introduction of innovations by firms modifies the structural conditions of the system. The changes in the structure of the system alter the conditions of all the other agents and in turn engender the introduction of new technologies.

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2.5. The early economics of knowledge

The economics of knowledge owes much to the classical legacy of Karl Marx (Rosenberg 1974). In the *Grundrisse* (Marx 1857–1858), Karl Marx's analysis of the central role of science in the capitalist process reveals extraordinary levels of clarity and insight. Technological change is fully endogenous to the economic system. More specifically, Marx argues that the levels of endogeneity of technological change are themselves an indicator of the advance of an economic system. The notion of knowledge as an endogenous productive force is clearly identified by Marx. Actually, the levels of endogeneity of knowledge, as a distributed economic force shared by a myriad of agents, fed by the combination of learning processes that lead to the accumulation of competence and tacit knowledge with scientific processes of deduction, and transformed into a means of accumulation of capital pushed by profit maximization (the general intellect), become a measure of the advance of capitalism as a social system.

The systematic application of scientific knowledge to the production process becomes, in Marx, the distinctive feature of capitalism. First, the collective character of technological knowledge is clearly grasped. Technological knowledge consists in a complex system of machines, skills and workers all characterized by distinctive elements of complementarity, interoperability and necessary compatibility. Second, technological and scientific knowledge are characterized by the strong elements of non-exhaustibility and limited appropriability. Their application, however, requires dedicated competence and resources that have a strong idiosyncratic character. In order to keep the process in motion, technological knowledge is constantly reproduced and expanded.

3. Innovation and competition – the Schumpeterian legacy

The legacy of Schumpeter (1912/1934) consists in a wide variety of contributions, from his first book, *The theory of economic development*, to the key article in the *Economic Journal* 'The instability of capitalism' (Schumpeter 1928), from *Business cycles* (Schumpeter 1939) and *Capitalism, socialism and democracy* (Schumpeter 1942) to the key contribution in the *Journal of Economic History* 'The creative response in economic history' (Schumpeter 1947).

The work of Joseph Schumpeter is at the heart of economics of innovation. He has provided the basic tools of the economics of innovation with the definition of innovation; the distinction between invention, innovation, imitation and diffusion; the understanding of the concentration of innovation in time and space with the notion of gales of innovations; the analysis of the key role of the corporation as the appropriate institution for fostering the rate of introduction of innovations; the understanding of the crucial role of credit to fund both the generation and the introduction of new technologies. Every student of economics of innovation is aware of the ground-breaking contribution of Schumpeter to the economics of innovation and technological change, thus an extensive and comprehensive analysis of his many contributions to this approach here would risk being repetitive.

Schumpeter, however, cannot be regarded only as the founding father of the economics of innovation, but also and mainly as one the key contributors to an economics of complexity where agents are credited with the actual competence to generate new knowledge and change their technologies. Technological change and economic change cannot be separated and they are the two inseparable features of the process of creative destruction that characterizes economic development. In Schumpeter, innovation, as opposed to invention, is the distinctive feature of the competitive process. Competition takes place by means of the introduction, adoption and diffusion of innovation, rather than by means of quantity and price adjustments. At the same time, competition among firms is the engine that drives firms to introduce innovations. By means of the introduction of new products, new processes, new organizations, new inputs and the identification of new markets, firms acquire a transient competitive advantage that feed extra profits. Eventually, their innovations will be imitated and perfect competition might be restored. Firms, however, can contrast the decline of profitability linked to the imitation of their innovations and the ensuing reduction of monopolistic power by means of the introduction of further innovations. Creative destruction is an endless process. In so doing, Schumpeter traced the basic guidelines to venture into an economics where change and novelty are the continual, and essential features of an endogenous process (Rosenberg 1994, 9–23).

Much attention has been given to the evolution of the thinking of Schumpeter on the drivers of technological change in economic development. A divide between the 'first' and the 'second' Schumpeter has been identified. The first Schumpeter, that is the tradition based on the *Theorie der Wirtschaflichen Entwicklung* originally published in German in 1911, pays attention to entrepreneurship as the driving mechanism. Entrepreneurs who create new firms to enter the markets are the primary source of technological innovations. The provision of credit by far-sighted bankers able to spot new ways of doing business is the complementary condition. The combination of entrepreneurs and far-sighted bankers is at the origin of the perpetual unrest of the economy. The key role attributed to entrepreneurship and its strong characterization in anthropological and sociological terms has raised some problems about the economic endogeneity of innovation. In the first Schumpeter entrepreneurs are indeed the essence of capitalist development but their emergence is mainly characterized as the result of sociological forces. The second Schumpeter is based on his 1942 book *Capitalism, socialism and democracy*. Here the driving role of the large corporation as the

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engine for the introduction of innovations is highlighted. The well-known Schumpeterian hypothesis is based on this second book: the monopolistic power, stemming from barriers to entry into existing product markets, of large corporations reduces the risks of imitation, feeds the accumulation of extra profits, hence increases the incentives to fund risky research activities, and favors the working of internal financial markets where the resources extracted by extra-profits can better match the competences of skilled managers. The large corporation appears as the institutional device that makes it possible to combine resources, incentives and competence to generating new technologies so as to increase the efficiency of the innovation process at the firm level and substitute external financial markets in the key role of the effective provision and correct allocation of funds to innovation with competent bureaucracies at the system level. A divide between static and dynamic efficiency arises: static inefficiency, stemming from monopolistic power, is compensated by dynamic efficiency, stemming from faster rates of introduction of new superior technologies. The second Schumpeter, however, also expresses some concern about the long-term viability of competitive mechanisms based on innovations due to the increasing routinization of activities leading to the introduction of innovations within the large corporation. Innovation here is fully endogenous, but 'routinized' within bureaucratic procedures. A large empirical literature has tested both hypotheses and proved their consistency, stressing the different role of large and small firms according to different types of industrial structure and technological regimes.

While there has been much debate on the differences highlighted in the evolution of Schumpeter's (1912/1934) thought from the entrepreneurial emphasis of his first book *The theory of economic development* to the view elaborated in *Capitalism, socialism and democracy* (Schumpeter 1942), the consistency of the theory implemented in the two key journal articles 'The instability of capitalism' of 1928, and 'The creative response in economic history' of 1947, deserves new attention and appreciation.

In 'The instability of capitalism', published in the *Economic Journal* in 1928, the 'two Schumpeters' are well integrated and coexist consistently. It thus seems appropriate to pay more attention to this contribution. Here, in fact, the theoretical distance between the dynamic analysis of the economic process, based on the understanding of the central role of technological change in market competition, and the Walrasian analysis of the general equilibrium, is especially clear. Innovation, as distinct from invention, is not only endogenous but the intrinsic element of the capitalistic economy. Innovation cannot be regarded as an external economy because this is the distinctive feature of the competitive process.

In his analysis of the role of creative reaction in economic history, Schumpeter (1947) fully elaborates the view that firms and agents in general are induced to react to the changing conditions of both product and factor markets in a creative way with the introduction of innovations, both in technologies and organizations, and by changing their products and processes. According to Schumpeter the distinction between adaptive and creative reactions is crucial to understand innovation as the source of growth and change. Firms and entrepreneurs are able to introduce innovations only when their reaction is 'creative' as opposed to 'adaptive':

> Whenever an economy or a sector of an economy adapts itself to a change in its data in the way that traditional theory describes . . . we may speak of the development as an *adaptive response*. And whenever the economy or an industry or some firms in an industry do something else, something that is outside the range of existing practice, we may speak of *creative response*. Creative response has at least three essential characteristics. First, from the standpoint of the observer who is in full possession of all relevant facts, it can always be understood *ex post*; but it can practically never be understood *ex ante*; that is to say, it cannot be predicted by applying the ordinary rules of inference from the pre-existing facts. This is why the 'how' in what has

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> been called the 'mechanisms' must be investigated in each case. Second, creative response … is an essential element in the historical process; no deterministic credo avails against this. Third, creative response – the frequency of its occurrence in a group, its intensity and success or failure – has obviously something, be that much or little, to do (a) with quality of the personnel available in a society, (b) with relative quality of personnel, that is, with quality available to a particular field of activity relative to the quality available, at the same time, to others, and (c) with individual decisions, actions, and patterns of behavior. (Schumpeter 1947, 149–50)

Schumpeter stresses the key role of the creative reactions and hence of innovations in the creation of radical dicontinuity in the path of economic growth:

> creative response … shapes the whole course of subsequent events and their 'long-run' outcome. It is not true that … [creative] responses dominate only what the economist loves to call 'transitions', leaving the ultimate outcome to be determined by the initial data. Creative response changes social and economic situations for good, or, to put it differently, it creates situations from which there is no bridge to those situations that might have emerged in the absence. (Schumpeter 1947, 150)

Finally, in the Schumpeterian approach, firms do more than adjusting prices to quantities and vice-versa: firms innovate. At the same time, innovation is no longer viewed as the result of the ingenuity of entrepreneurs only (outsiders who enter the marketplace by means of new products or processes): incumbents innovate as well, and do so in order to face unexpected changes in the economic environment. With the notion of creative response, innovation becomes fully endogenous to the economic system. The dynamic efficiency of a system, as measured by the capability to increase its overall efficiency by means of the introduction of technological and organizational innovations, becomes the new key parameter to assess actual welfare.

In this context, the analysis carried out in *Business cycles* (Schumpeter 1939), where the collective character of the innovation process, the interdependence among innovators, and the complementarity of the new technologies within the gales of innovations are highlighted, and the intertwined co-evolution of economic institutions, industrial structures, economic architectures of interactions and exchanges and consumers' preferences along with the innovative process is historically substantiated, paves the way to a richer understanding of the systemic and inherently complex character of the innovation dynamics.

3.1. Innovation and entrepreneurship

Following Schumpeter Mark 1 – the literature inspired by *The theory of economic development* – the supply of entrepreneurs able to spot new technological opportunities and to understand the possible technological and economic applications of new scientific breakthrough is considered an important factor in understanding the pace of introduction of new technologies and their specific economic and technological characteristics. This approach praises the role of new firms as vectors of new technologies and suggests that only high birth levels of new firms can sustain rates of technological change. Strong empirical evidence has been provided by Acs and Audretsch (1988) to support the hypothesis.

Analysis of the institutional and economic conditions that favor entrepreneurship, and the entry of new innovative firms in the marketplace generally, becomes an important area of investigation. Entrepreneurship in this context supplies evidence of the key role of meta-economic factors in assessing the rate and direction of technological change.

Expanding this line of enquiry, Baumol (2004) has highlighted the role of the social organization of economic, institutional and social mechanisms of identification and the

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valorization of the 'given' supply of creative talents distributed at random in any economic system. The larger the number of creative talents each system is able to identify and valorize, the larger is the dynamics of growth in output and efficiency in the economic system. Here creative talents are an exogenous characteristic distributed at random, but the filtering mechanisms elaborated within the economic system are endogenous.

3.2. Innovation and the corporation

The so-called Schumpeterian hypothesis sketched in *Capitalism, socialism and democracy* articulates the approach that large firms are necessary for high rates of technological advance to take place. Barriers to entry and monopolistic competition provide to corporations *ex ante* appropriability, reducing the risks of leakage and imitation. In turn, large price-cost margins and competence provide corporations with the opportunity to match internal financial resources with dedicated information and competent decision-making so as to fund new and promising research activities. Here, the large firm takes on a central role and appears the locus of accumulation of sticky technical knowledge and hence technological progress. As Nelson (1959), shows, the funding and performance of research and development (R&D) activities become an integral part of the conduct of large corporations.

In this context, Chandler (1977, 1990) provides the foundations of the resource-based theory of the firm with emphasis placed on the characteristics of the corporation as the institutional locus for fostering the accumulation of technological knowledge and economic competence.

A rich empirical literature investigates the relations between the characteristics of corporations, their conduct, their innovation strategies and their performances. Cantwell and Andersen (1996) provide evidence on the extent to which technological leadership lasts over time. Haneda and Odagiri (1998) confirm that technological assets are positively related to the ratio of the stock exchange value to the book value of corporations (the so called Tobin's q).

3.3. Innovation and market structure

The Schumpeterian approach to innovation as an essential component of the competitive process is consistent with the Marshallian interpretation of the competitive process. Variety and selection are essential elements of the Marshallian notion of competition. Firms, diverse in terms of size, location and efficiency, confront each other in the product marketplace and are sorted out by the workings of the competitive process. Entry and exit feed the dynamics of the process. Here, each firm is confronted with a continual redefinition of its relative market context and must face the competitive threat brought about by firms that are able to produce at lower costs, either because of access to cheaper production factors or more effective production technologies. In the Marshallian competition, the duration of the adjustment process to an eventual equilibrium is endless, and firms experience prolonged out-of-equilibrium conditions in which they can earn transient yet heterogeneous profit levels.

The Schumpeterian legacy has been especially fertile in articulating the key relationship between rivalry and intentional innovation, and has allowed exploration of the causal relations between barriers to entry, levels of markups, market structure and incentives to introduce new technologies as articulated by Scherer (1967). Scherer (1970) implements the structure–conduct–performance framework, drawn from industrial organization, within the notions of technological opportunities – defined as the opportunity to introduce

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technological innovations impinging on exogenous scientific breakthrough – as a key aspect of the industrial structure, and innovation as one of the main conducts of firms. The enriched structure–conduct–performance framework has made it possible to gather empirical evidence that confirms that the size distribution of firms, the levels of concentration and the forms of competition among firms do indeed affect the rates of introduction of innovations and their characteristics. The rich survey by Kamien and Schwartz (1975) provides a detailed review of this literature.

Dasgupta and Stiglitz (1980) provide an original framework to understand how the tools of oligopolistic rivalry make it possible to identify the 'equilibrium' amount of R&D expenditure, within an industry, but fail to grasp the determinants of the actual amount of innovations that R&D expenditures can yield. This line of enquiry fails to appreciate the key elements of the aggregate dynamics by means of which individual research efforts interact and eventually translate into innovation. A substantial gap emerges between the understanding of the amount of research efforts and the identification of the actual amount of innovations introduced and put in use, respectively, at the firm and the system level.

Aghion and Howitt (1992) elaborate the Schumpeterian notion of creative destruction with a model based upon a vertical quality ladder. Innovations consist in new intermediary inputs. Monopolistic competition characterizes the markets for such production factors. Innovators do enjoy short-term monopolistic profits. Profits, however, are inherently transient. New technologies, in fact, destroy the competitive advantage of the previous generations of innovators. The ingenuity of the model consists in the identification of an inter-temporal trade-off between present levels of research and future levels of profits. At each point in time, there should be a direct and negative relationship between the current levels of research funded by firms and the future levels of monopoly profits. The profitability of innovations affects directly their price. In this context, an equilibrium solution can be found where the derived demand for innovations matches their supply. Hence the model is able to endogenize the levels of R&D expenditures and the rates of introduction of innovations. The effects of the flow of the pecuniary externalities that stem from the provision of new inputs that embody technological innovations to downstream sectors explain the increase of total factor productivity growth at the system level.

The vertical analysis of the creative destruction can be further enriched when the cascade effects of the introduction of innovations in upstream industries in terms of induced innovations and creative adoptions in downstream sectors is appreciated (Antonelli 2008c).

The study of the processes of the delayed adoption of innovations, i.e. of the diffusion of innovations, rather than of their early introduction, provides a rich and fertile context of empirical and theoretical application of the Schumpeterian framework. The time patterns of entry of firms and hence the evolution of industrial demography, concentration, profitability and rates of growth of both firms and industries is analyzed within the Schumpeterian sequence of early monopolistic power followed by imitative entry and finally competition. Diffusion matters on both the demand and the supply side. As Stoneman and Ireland (1983) show, new technologies are adopted only if and when they fit specific product and factor markets conditions: some agents will never adopt a new technology and the identification of the determinants of such non-adoption becomes relevant. Adopters are no longer viewed as passive and reluctant perspective users, but rather as ingenious screeners who assess the scope for complementarity and cumulability of each new technology with their own specific needs and contexts of action. Profitability of adoption is the result of a process rather than a given fact. Technology diffuses when it is applied to diverse conditions of use. The intrinsic heterogeneity of agents applies in fact not only to their own technological base, but also to the product and factor markets in which they operate.

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The vintage structure of fixed costs and tangible and intangible capital is portrayed, by Antonelli (1993) as major factors of differentiation and identification of the specific context of action, with respect to both technological change and market strategy. The flow of investments is determinant in assessing the rates of adoption: firms and countries with low investment rates have fewer chances to adopt new capital goods. Lower levels of penetration of new capital goods reduce the competitive edge of firms and hence their rates of growth and subsequently their rates of investment. In this case, typical positive feedback occurs: fast rates of investment favor the diffusion of innovations that feed faster rates of growth and hence higher rates of investment. The economics of complexity sneaks in.

Katz and Shapiro (1986) stress the role of the changing conditions of usage in the diffusion of innovations. Important changes in the profitability of adoption may take place on the demand side because of the effect of network externalities, i.e. the preference of consumers for a new good is influenced by the number of adopters.

Within the context of the economics of localized technological change, Antonelli (2006a, 2006b) blurs the distinction between innovation and diffusion: adoption is viewed as a complementary component of a broader process of adjusting the technology when unexpected events in the product and factor markets push firms toward a creative reaction. When the stock of adoptions exerts a suitable combined effect on the gross profitability of adoption and on the costs of adoption, such that the net profitability of adoption and hence the rates of new adoption follow a quadratic path, the dynamics of creative adoption can engender an *s*-shaped diffusion process.

Pavitt (1984) accommodates and operationalizes the variety of paths of technological change across sectors and technologies. In so doing, he implements the neo-Schumpeterian structure–conduct–performance approach elaborated by Scherer and paves the way to the notion of 'technological regimes', implemented by Malerba and Orsenigo (1995) so as to further expand the notion of industrial structure and include the characteristics of knowledge in terms of appropriability and cumulability. Similar efforts are made by Klepper to update the notion of product cycle and include the rates of entry and exit and the critical phases of industrial shake-out that follow the introduction of an innovation. Incumbent innovators can take advantage of previous innovations in many ways: the early competitive advantage makes it possible to fund new research; acquired competence and technological knowledge are useful inputs for further innovations; barriers to entry based on market shares and size delay imitation; and technological advance feeds diversification and entry in new industries (Klepper and Graddy 1990).

An important contribution is made with the notion of dominant design elaborated in by Abernathy and Clark (1985). The dominant design is the result of a new interpretation of the innovation process. It provides an alternative route to the traditional sequence between the transient monopolistic power stemming from the introduction of an innovation, eventually followed by a few imitators leading to oligopoly and monopolistic competition. After the introduction of a variety of rival technologies by many rival firms, a selection process takes place and a few leading firms, able to elaborate the dominant design, emerge out of the competition with a consistent competitive advantage. Monopolistic rents emerge at the end of the selection process and may last. The diffusion of a new technology is no longer seen as the outcome of the adaptive adoption of a new single technology, but rather of the choice of one new technology among many. Diffusion is the result of the selection of a dominant design out of an original variety of different technological options. New ideas can be implemented and incrementally enriched, so as to eventually become profitable innovations, only when appropriate coalitions of heterogeneous firms are formed both on the demand and the supply side.

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Technological choices concerning the introduction of product and process innovations, the adoption of new technologies provided by suppliers and the imitation of competitors are mingled with market strategies, such as specialization, outsourcing, diversification, entry and exit, merger and acquisitions and internal growth. In a continual trial in the marketplace, firms experiment their changing mix of technological and market conduct. At the aggregate level, the result is the market selection of new and better technologies, often characterized by strong systemic complementarities. As Henderson and Clark (1990) suggest the structure of interactions and the changing coalitions between different groups of players in overlapping, yet specific technological arenas shape the rate and direction of technological change in general. In this context, the shake-out of the system may favor the emergence of new industrial architectures where firms and technologies find a new contextual location.

The notion of general-purpose technology introduced by Bresnahan and Traitenberg (1995) and further elaborated by Lypsey, Bekar, and Carlaw (1998) stresses the systemic character of technological change. New general-purpose technologies, are the result of the complementarity and interdependence of a variety of technological innovations being sequentially introduced, and are characterized by high levels of fungibility as they can be applied to a great variety of production processes.

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4. Innovation and knowledge: the Arrovian legacy

The Arrovian legacy explores the many facets of the economic analysis of knowledge as an economic good, both as an input or a production factor and an output of an intentional process of creation.

Arrow (1962b) makes an important contribution, focusing on the analysis of knowledge as an economic good *per se*, no longer embedded either in capital products or organizations. Prior to this, knowledge was not regarded as a separate item. The analysis of knowledge as an economic good made it possible to grasp the causes of the radical failure of the marketplace to perform its traditional functions and the ensuing risks of the underproduction of knowledge in market systems.

The results of the new approach are ground breaking. Knowledge is the basic intermediary input for the increase of efficiency, and the incentive, in terms of social desirability to the production of knowledge, is huge, as any economic system would dedicate most of its resources to the generation of new knowledge as a way of increasing efficiency in the production of all the other goods. However, because of the major limitations of knowledge as an economic good in terms of non-appropriability, non-excludability, non-rivalry in use, non-exhaustibility and non-divisibility, the private profitability of knowledge-generating activities is well below social desirability. Moreover, because of the high levels of uncertainty both in generation and appropriation, economic systems are unable to fund the correct amount of resources for the generation of new knowledge and therefore to increase the production of goods via the increase in the general efficiency of the production process. Hence dynamic inefficiency adds to static efficiency: the markets for knowledge, as a standalone good, are inefficient and therefore the necessary levels of division of labor and specialization cannot be achieved. A radical market failure is the direct consequence of the characteristics of knowledge, as an economic, private and unbundled good. The failure of markets for knowledge is twofold: it takes place in the markets for knowledge as an output, and in the markets for financial resources necessary to undertake its generation.

Dasgupta and David (1994) implement the attribution to scientific knowledge of the characteristics of a public good and legitimate a division of labor between firms and

universities. The latter are responsible for the production and distribution of scientific knowledge as a public good. The State's role in this situation was that of an indispensable intermediary that collected the taxes necessary to finance university research. Scientific inventions are mainly perfected and improved in the academic institution. The university is a well-defined institution that has emerged through centuries of implementation. Scientific knowledge therefore is the output of a highly idiosyncratic ethos and dedicated incentives where reputation-seeking behavior substitutes standard profit-maximization. Scientific knowledge produces effects in terms of technological opportunities. Firms in fact are expected to be able to collect and implement the stimulus that was set off by new scientific discoveries. Firms will eventually take advantage of the provision of new scientific breakthrough and use them as the basis for the actual generation of technological innovations.

Pakes and Griliches (1980) confirm that, at the firm level, technological knowledge, as measured by patent statistics, can be considered an output in a knowledge-production function where R&D activities are the main inputs. The empirical evidence gathered and analyzed by Mohnen, Mairesse, and Dagenais (2006) confirms, however, that the local context of activity has strong influences on the effects of R&D activities.

Jaffe (1986), Feldman (1999) and Griliches (1992) explore the key role of other – than the traditional R&D expenditures – and yet relevant inputs in the knowledge-production function. They identify the contribution of external knowledge that spills from the research efforts of other firms and public research laboratories and can be used in the production of new knowledge by other parties. The rediscovery of knowledge spillovers leads to the revival of the Marshallian analysis of externalities. Regional economics contribute to the analysis of spillovers with substantial understanding about the role of geographic proximity in favoring access to external knowledge with positive effects on the productivity of resources invested internally in R&D expenditures (Audretsch and Feldman 1996).

The Arrovian view of knowledge as a quasi-public good is contrasted by a radical change in perspective that stresses the role of learning by doing and learning by using as the basic engine of accumulation of knowledge. New technological knowledge stems from such learning processes and especially from efforts to convert tacit knowledge into new procedures that can be partly shared and transferred (Stiglitz 1987). The new bottom-up understanding of the discovery process contrasts the traditional top-down approach to the origin of technological innovations. The analysis of the accumulation of technological knowledge plays a key role in this context. Malerba (1992) highlights the importance of tacit knowledge, embedded in the organization of innovators and especially in their learning procedures, to reduce the capability of perspective imitators to absorb the new knowledge and to favor higher levels of appropriability.

4.1. Information economics for the economics of knowledge

The distinction introduced by Arrow (1969) between information economics and economics of knowledge provides basic guidance to implement the economics of knowledge. An array of tools elaborated by the economics of information – such as agency theory, transaction costs analysis, signaling theory and economics of contracts – is applied successfully to understand the generation and dissemination and the use of knowledge.

The costs of imitation and absorption of external knowledge are gradually identified. Mansfield, Schwartz, and Wagner (1981) explore empirically the notion of knowledge appropriability. Cohen and Levinthal (1990) elaborate the key notion of absorption costs

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and stress that an active role of users and imitators is necessary in order to actually take advantage of the potential benefits of knowledge spillovers.

Nelson (1982) shows that, next to the amount of resources invested in the production of knowledge, the efficiency of the internal production of knowledge is a central issue and highlights the key role of learning processes and, most important of external knowledge as a source of inputs.

In this second step, the debate shifts toward the basic issue of the intrinsic complementarity and interdependence, at the technological, industrial and regional levels, among the agents in the accumulation of new technological knowledge and economic competence and subsequently in the introduction and adoption of new technologies. Knowledge is now viewed both as the output of a specific research and learning process, and as the input for other activities leading to the generation of new knowledge. Here again the dynamics of positive feedback is at work: the output of one part of the system is the input for another, and yet such interactions are mediated by the price mechanism only to a limited extent – complexity economics is again closer.

Hayek (1945) elaborates the notion of distributed knowledge, dispersed and fragmented in a myriad of economic agents and provides the foundations for the new understanding. Only when a complementary set of knowledge fragments is brought together within a context of consistent interactions, can successful innovations be introduced and adopted: technological knowledge is the product of a collective activity. Von Hippel (1976, 1998) paves the way to an array of empirical analyses on the key role of user–producers interactions as basic engines for the accumulation of new technological knowledge and the eventual introduction of new technologies are important at this stage. At the same time, the appreciation of the key role of internal learning leads to see how technological knowledge is characterized by considerable stickiness (Lundvall 1988).

Teece (1986) shows how the basic tools of information economics, such as the analysis of transaction, agency and communication costs, provide basic guidance to elaborate an integrated framework able to understand the matching between types of knowledge and modes and mechanisms of knowledge governance. In so doing, David Teece provides new foundations to the Schumpeterian analysis of embodied knowledge and contrasts the Arrovian analysis of knowledge as a good *per se*. Along these lines, March (1991) introduces the fertile distinction between knowledge generation and knowledge exploitation.

Arora and Gambardella (1994) provide empirical evidence on the array of institutional arrangements that make possible knowledge transactions calling attention on the enforcement of appropriate contracts, exchanges of hostages within technological clubs and long-term, repeated interactions. Patrucco (2005) shows that communication costs play a key role in assessing the actual capability of firms to access relevant external knowledge and contribute the emergence of new technological systems. Proximity of firms to universities and public research centers in general becomes a major source of access to external knowledge provided some effort is made to absorb the knowledge available in the local knowledge commons.

The empirical analyses reviewed by Antonelli (2006b, 2006b) within the frame of the economics of information, show that technological knowledge is a highly heterogeneous dynamic process characterized by varying levels of appropriability, tacitness and indivisibility that take the forms of cumulability, complexity, fungibility and stickiness. The heterogeneity of knowledge leads to different modes of knowledge governance, articulated in a variety of hybrid forms ranging from coordinated transactions and constructed interactions to quasi-hierarchies, which can be found between the two unrealistic extremes of pure markets and pure organizations.

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4.2. The new growth theory

The new growth theory builds on three important acquisitions of economics of knowledge: (a) the distinction between generic and tacit knowledge, and the related notion of technological knowledge as a quasi-public good because of quasi-appropriability; (b) the understanding of technological externalities and the dynamics of spillover; and (c) the notion of monopolistic competition as a result of the introduction of new products.

According to the analysis of Romer (1994), economic growth relies on collective access to generic knowledge, which flows in the air. Romer distinguishes between generic technological knowledge, which is germane to a variety of uses, and specific technological knowledge that is embodied in products and as such has strong idiosyncratic features. Specific knowledge can be appropriated; generic knowledge instead retains the typical features of the Arrovian public good. Innovators generate generic knowledge while engaged in the introduction of new specific knowledge embodied in new products and new processes. The production of specific knowledge takes advantage of the collective availability of generic knowledge. The spillover of generic knowledge helps the generation of new specific knowledge by third parties and yet does not reduce the incentives to the generation of new knowledge for the strong appropriability of the specific applications.

While the new growth theory has been able to adopt and adapt to much of the progress put forward by the economics of innovation, it misses the core of the analysis. The actual determinants of the partial excludability and appropriability of the new knowledge and, more specifically, the ratio of the generic component of the new knowledge, with respect to the idiosyncratic one, are not investigated. Hence the definition of the incentives to the generation of new knowledge on the one hand, and the actual contribution of the spillovers of generic knowledge to the generation of new knowledge, on the other, remains unclear (Adams 2006).

4.3. Knowledge as the product of systemic interactions

The contributions of Freeman (1991), Patel and Pavitt (1994), stress the systemic features of the process of generation and usage of technological knowledge show how knowledge complementarity, weak divisibility and interdependence among learning agents, firms and public research centers become central to understanding the attributes of specific national innovation systems articulated in technological, industrial and regional subsystems characterized by networks of interaction and communication into which the dissemination and access to technological knowledge takes place.

The generation and introduction of technological innovations are now viewed as the result of complex alliances and compromises among heterogeneous groups of agents. Agents are diverse because of the variety of competencies and localized kinds of knowledge they build on. Alliances are based on the valorization of weak knowledge indivisibilities and local complementarities among technological different kinds of knowledge (Arora 1995; Link and Scott 2002). Trajtenberg, Henderson and Jaffe (1995) confirm that the convergence of the efforts of a variety of innovators, including corporations and universities, each of which has a specific and yet complementary technological base, can lead to the successful generation of a new technology. Here the Schumpeterian notion of gales of innovations that characterize business cycles revives and reveals its heuristic strength (Schumpeter 1939).

Once more the dynamics of positive feedback emerges as the key factor. Now, however, it is clear that positive feedback can take place only when an appropriate architecture

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of network relations is formed. James Adams highlights the critical role played by the architecture of network relations especially between corporations and public research centers marks an important step toward the foray of complexity economics.

The new understanding about the asymmetry between debt and equity in the provision of funds for research activities paves the way to a revolution in financial markets. Equity finance has an important advantage over debt in the provision of funds for innovative undertakings because it can participate within the bottom tail of the highly skewed distribution of positive returns stemming from the generation of new knowledge and the introduction of new technologies. This has important consequences both in terms of the reduction of both the risks of credit rationing and the costs of financial resources for research activities. Lenders, in fact, need to charge high interest rates in order to compensate for the risks of failure and to sort out a large portion of new research activities to avoid as many 'lemons' as possible. Equity investors instead find an equilibrium rate of return at much lower levels because they can participate within the huge profits of a small fraction of the new ventures. As Gompers and Lerner (2001) argue the fraction of lemons that equity can support is much larger than that of debt; as a consequence, financial equity by means of venture capital can provide a much larger amount of funding for research activities.

The creation of technological platforms centered on new key technologies by means of the cooperation of rival innovators favors upstream the convergence of technologies, and increases downstream the scope for both the widespread diffusion of applications and the introduction of incremental enrichments. This type of systemic approach to innovation reappraises the relationship between services and manufacturing as complementary activities, and overcomes previous understanding rooted in the divide between the two.

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5. Evolution and complexity

This approach is the result of the integration within the new emerging paradigm of complexity of three different strands of analysis: (a) the early biological grafting; (b) the analysis of learning; (c) the new epistemology based on notions of tacit knowledge, bounded and procedural rationality and the behavioral theory of economic action (March 1988).

Building upon the Marshallian understanding of the economic system as an organism with strong similarities to biological entities, these different trails converge toward the notion of innovation as an emergent property of a complex evolving system. In this approach, innovation is the result of a path-dependent and collective process that takes place in a localized context if, when and where a sufficient number of failure-induced creative reactions are made in a coherent, complementary and consistent way (Marshall 1890).

The introduction of the analysis of partial equilibrium enables Marshall to blend the basic microeconomics of the neoclassical approach and the dynamic approach of Marx and Smith. Marshall makes explicit that knowledge is a key component of capital and itself a production factor and identifies the collective character of the generation of technological knowledge as an endogenous process where a variety of agents, co-localized within the industrial districts, contribute complementary bits of knowledge. Knowledge externalities play a key role in providing firms with essential inputs for the generation of new knowledge. Moreover, Marshall accommodates, within competitive markets, the heterogeneity of firms and explains this in terms of the different levels of knowledge and competence possessed by each firm. Quasi rents are the direct remuneration of the stock of knowledge and competence that each firm has been able to accumulate and valorize. These steps push Marshall to appreciate the role of biology as the Mecca of economics.

Indeed, biology provided important suggestions and stimulation to the early economics of innovation.

In *Principles*, Marshall is able to blend the legacy of Smith with the neoclassical approach, grasping the dynamic complexity of structural change, as articulated in the interaction between specialization and technological change leading to a growing heterogeneity of firms in a context characterized by variety and complementarity:

> The development of the organism, whether social or physical, involves an increasing subdivision of functions between its separate parts on the one hand, and on the other a more intimate connection between them. Each part gets to be less and less self-sufficient, to depend for its well-being more and more on other parts, so that any disorder in any part of a highly developed organism will affect other parts also. This increased subdivision of functions, or 'differentiation,' as it is called, manifests itself with regard to industry in such forms as the division of labor, and the development of specialized skill, knowledge and machinery: while 'integration,' that is, a growing intimacy and firmness of the connections between the separate parts of the industrial organism, shows itself in such forms as the increase of security of commercial credit, and of the means and habits of communication by sea and road, by railway and telegraph, by post and printing press. (1890, Book IV, Chapter VIII, Sections 2, 3 and 4, 241)

5.1. Biological grafting

A first relevant basket of important research programs favored by biological grafting is the analysis of the delays in the adoption of given technological innovations. The economics of the diffusion of new technologies is conceived of as the study of the factors that account for the distribution over time of the adoption of identifiable successful innovations. A new technology is introduced after a scientific breakthrough and yet it takes time for all perspective users to adopt it. The successful and still widening application of the epidemic methodology emerges in this context. The time distribution of adoptions can be conceived of as a result of the spread of the contagious information about the profitability of the new technology. Proximity in geographical, industrial and technical space matters here in that it provides reluctant and skeptical, risk-adverse adopters with the opportunity to assess the actual profitability of the new technology and hence to adopt it. Griliches (1957) grafts epidemiology into economics of innovation, providing a new analytical tool and a fertile context for empirical analysis where contagion is assimilated to diffusion assuming that agents are heterogeneous, and the engine of the dynamics is based upon the spread of information.

Metcalfe (1981) provides a significant improvement to epidemic diffusion: next to epidemic contagion on the demand side, changes in supply also account for the distribution over time of adoptions. In so doing, Metcalfe reintroduces the basic laws of standard economics into the epidemic framework and shows the relevance of their dynamic interplay. A sequence of logistic diffusion paths takes place when relevant changes on the supply side affect the spread of the epidemic contagion in new categories of perspective adopters.

The second relevant biological graft into economics of innovation is provided by the life cycle metaphor. The life cycle metaphor has been around in the theory of the firm ever since the Marshallian forest's trees. A shift takes place when the sequence of birth, adolescence, maturity and obsolescence is applied to framing the steps in the life of a new product instead of a new firm. After introduction, the life of new products is characterized by a number of systematic events. According to the product life cycle approach, a consistent pattern can be identified in the typology of innovations being introduced, in the evolution of the demand, in the industrial dynamics and in the characteristics of the growth of the firm.

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Utterback and Abernathy (1975) apply the life cycle metaphor to technological innovations: (a) the distinction between major innovations and minor ones is articulated and a sequence is identified between the introduction of a major innovation and the eventual swarm of minor, incremental ones; (b) a sequence between product and process innovations is identified. After the introduction of a new product, much research takes place in the effort to improve the production process.

Soete and Turner (1984) explore the details of the selection process in biology and apply the notion of replicator to grasping the sequential features of industrial dynamics along the trajectory. Metcalfe (1997) has shown the fertility of the replicator in understanding how innovators can earn extra profits, fund their growth and acquire larger market shares. The analysis of the diffusion of innovation is intertwined with the study of the selection mechanism in the marketplace. Firms that have been able to introduce new technologies are also able to increase their growth and their market shares.

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5.2. Learning as the engine of growth

The economics of learning makes a major contribution to the understanding of the dynamics of technological change. Penrose (1959) elaborates the notion of learning as the engine of growth and qualifies the firm, its organization and its routines, as the privileged actor in the learning process. The firm precedes the production function as its primary activity consists of the generation of new technological knowledge. Each firm, as it is well known, learns and builds up new capabilities and eventually discovers new possible applications for production factors and competences that are found within its own boundaries. According to Edith Penrose, in other words, innovative firms are successful when they try and make the most effective use of production factors that are not only locally abundant, but also internally – within its own boundaries – abundant. Edith Penrose stresses the strategic efforts of corporations to exploit technological knowledge generated by means of systematic learning processes with the identification, creation and valorization of idiosyncratic production factors.

Arrow (1962a) laid the foundations for a theory of economic growth based on learning processes that make possible the generation of new knowledge and, eventually, the introduction of new technologies. Agents, as well as firms, are now credited with the capability to learn. Learning is the result of repeated actions over time and reflective thinking. Learning has strong cumulative features and as such leads to dynamic increasing returns where cost reduction is associated with time rather than sheer size of production.

Learning is consistent with general equilibrium analysis only as long as it applies to the representative agent: learning should be ubiquitous and symmetric across agents in the system. The evidence, however, shows that the distribution of the residual is highly uneven across regions, industries, firms and, especially, historic phases. The analysis of learning processes make it possible to appreciate the role of a variety of contextual constraints in shaping the process: learning is possible only in the limited spectrum of techniques and conditions, both internal and external to the firm. Hence technological change is localized.

The analysis of learning, in fact, has been much stretched and sharpened by the insight of Atkinson and Stiglitz (1969) who elaborated further upon the key role of learning in the generation of new technologies and introduced the strong hypothesis that technological change can take place only in a limited technical space, defined in terms of factor intensity. Technological change is localized because it has limited externalities and affects only a limited span of the techniques, contained by a given isoquant, that are identified by the actual context of learning. In other words, technological change can only take place where

firms have been able to learn: the localization here is strictly defined in terms of factor intensity and with respect to the techniques in place at each point in time.

From this viewpoint, the resource-based theory of the firm, as elaborated upon the contributions of Penrose (1959) and the analysis of Atkinson and Stiglitz (1969) can be considered together as sequential steps in a process of specification and incremental enrichment of the identification and analysis of the implications of the economics of learning, as an out-of-equilibrium process.

Developing the localized technological change approach, Antonelli (1995, 1999, 2008b) argues that because there are irreversibilities, limited knowledge and local learning, the introduction of new technologies is induced by the disequilibrium conditions brought about in each system by all changes in product and factor markets. Hence the direction of technological change in terms of its specific form of bias and how it is introduced and adopted, however, reflects the specific conditions of local factor markets. Well-defined long-term technological paths emerge in each region and depend on the selection process in product markets. The more rigid and idiosyncratic, hence irreversible, both the capital structure, tangible and intangible specific to each firm and the endowment of production factors and the system of relative prices, the more specific the technological path of each region is likely to be (Arrow 2004).

In the localized technological change approach, innovations are not the automatic result of learning without any intentional and explicit effort. The inducement context, characteristic of the localized technological change approach, makes it possible to overcome this major limitation. The analysis of the specific context into which learning takes place, provides opportunity for the analysis of the intentional action of generation of new technological knowledge and introduction of new technologies.

The bottom-up approach to understanding the dynamics of knowledge finds here the first input, and in so doing, it stresses the role of technological knowledge, acquired by means of localized learning processes, as the primary input in the generation of new knowledge at large, together with the scientific advances made available by the scientific community and the acquisition of external knowledge spilling in the atmosphere.

The analysis of the role of the structural characteristics of economic systems in general, and specifically of the role of the structure of relative prices, as determined by the endowment of basic inputs, and of the dynamics of industries and sectors as factors shaping the rate and direction of technological change, provides a historical context into which the analysis of the interplay between technological and structural change makes a step forward.

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5.3. Rationality and change

Two important contributions drawn directly from the philosophy of science and early cognitive science characterize the emergence of technological trajectories as the new heuristic metaphor and a research agenda. The distinctions introduced between tacit and codified knowledge and bounded and procedural rationality, as opposed to Olympian rationality, can be considered the founding blocks.

According to Polanyi (1969): 'Knowledge is an activity which would be better described as a process of knowing' (Polanyi 1969, 132), rather than a good, and agents often know more than they are able to spell in a codified and explicit way. Tacit knowledge is embedded in the idiosyncratic procedures and habits elaborated by each agent. It can be translated into a fully codified knowledge only by means of systematic and explicit efforts. An important implication of the distinction between tacit and codified knowledge consists in fact in the increase in the 'natural' appropriability of technological knowledge.

Simon (1979) makes two important contributions here. In the first, he introduces the notion of bounded rationality in order to stress the limitations of traditional assumptions about the Olympian rationality of '*homo oeconomicus*'. Bounded rationality quickly became a building block for the new emerging economics of information: the acquisition of information and the generation of signals are costly and economics needs to care about them.

Subsequently, however, Simon elaborated the distinction between substantive and procedural rationality. The introduction of the notion of procedural rationality has far-reaching consequences as it introduces the notion of sequential decision-making. Agents cannot achieve substantive rationality for the burden of the wide range of activities necessary for gathering and processing all the relevant information. Agents can elaborate procedures to evaluate at each point in time and space the possible outcomes of their behavior, but within the boundaries of a limited knowledge and using satisfying criteria as opposed to maximization rules.

The notion of procedural rationality introduced by Simon marks a major contribution to the economics of innovation. Olympian rationality is at odds with a context characterized by radical uncertainty where nobody actually knows the outcome of a research project and even less so the next direction of the technological changes being introduced. Indeed, the very notions of future prices and future markets cannot even be considered when technological change is taken into account. In such a context, only sequential decision-making based on limited information and limited knowledge is possible.

A behavioral theory of economic action is implemented upon these foundations. According to James March (March and Simon 1958; Cyert and March 1963), because firms rely upon procedural, as opposed to substantive rationality, decision-making is based upon satisfying procedures and the identification of satisfactory levels of performances. Firms are risk adverse and hence reluctant to change: they overcome intrinsic inertia and resistance to change only when the actual levels of performance fall below some thresholds. At this time, a failure-induced mechanism of change is set for and firms are more ready to take the risks associated with innovation (March and Shapira 1987). In the behavioral theory of the firm, innovation is an out-of-equilibrium conduct that takes place when out-of-equilibrium conditions prevail. In equilibrium conditions, the firms are reluctant to innovate because of the intrinsic uncertainty that characterizes both the generation and exploitation of technological knowledge. Innovation is induced by the failure to reach the levels of performance that each firm has defined as satisfactory.

The application of the notion of procedural rationality to economics of innovation leads to a new appraisal of the role of the creative reaction as the new understanding of the basic inducement of innovation. Antonelli (1989) implements and elaborates the Schumpeterian notion of creative reaction as the qualifying aspect of the behavior of innovative agents: agents innovate when their expectations are deceived and their performances fall below subjective levels of aspiration. Creative reaction is a part of the satisfying behavior of economic agents afflicted by bounded rationality but able to rely on their procedural rationality and to learn, and hence to generate new knowledge and to modify their conditions.

5.4. Evolutionary approaches: routines and trajectories

The real problem facing economics of innovation is to provide an economic context within which to understand the behavior of economic agents facing radical uncertainty and the multiple possible outcomes of their choices. A broader notion of rationality is needed as well as a more articulated understanding of the complexity of social interactions, beyond

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the standard price-quantities adjustments selected in a context of perfect foresight. Nelson and Winter (1977) apply fruitfully the notions of tacit and codified knowledge and the implications of bounded and limited rationality to the theory of the firm with the notion of routines. Loasby (1998) argues that routines provide a clue to understanding growth as the result of the innovative behavior of myopic agents able to accumulate knowledge and to convert it into competence.

Nelson and Winter (1975, 1982) are able to combine the notion of routines with the Schumpeterian notion of innovation as the basic competitive tool and elaborate the notion of technological trajectories, defined as the sequence of innovations aligned along a cumulative learning process. The notion of trajectory makes it possible to mimic, with simulation techniques, the working of a system where myopic firms follow innovative routines in order to compete along technological trajectories, and in so doing generate growth.

Dosi (1982) implements the notion of trajectories and applies it both to understanding the dynamics of innovation with respect to the sequence of well-defined technologies and to the sequence of innovations introduced by well-identified firms and, eventually, economic systems such as regions, industries and even countries. The analysis of trajectories appears especially promising at the firm level and in the analysis of the competitive process. First-comers reap substantial competitive advantages and build barriers to entry based on their technological knowledge. Long-lasting extraprofits provide financial resources to fund the incremental implementation of internal learning by doing and accumulated competence.

The notion of technological trajectory builds on the achievements made in terms of the product life cycle, and makes possible a spring of cumulative research in the discipline, including the notion of technological convergence introduced by Rosenberg (1963) to stress the dynamic blending of technologies and their generative relations.

Empirical research makes it possible to identify a variety of trajectories. When a variety of trajectories in a variety of technologies and firms is identified, a number of basic questions arise: why some trajectories are 'steeper' than others; why some trajectories 'last' longer than others; why some firms fail to innovate; and why some industries are less able than others to build their own trajectories. Silverberg, Dosi, and Orsenigo (1988) elaborate the analysis of the possible outcome stemming from multiple interactions among a variety of trajectories applied both to firms and technologies.

Agreement on the trajectory metaphor disappears quite rapidly when its strong deterministic bent is fully revealed. This trajectory metaphor seems to revive the old temptation to use *ad hoc* technological determinism to explain social and economic changes as a process of sequential alignment dictated by technology. Paradigmatic crises arise as factors of discontinuity. New trajectories are generated and old ones decline. The origin of such changes and the emergence of new technological paradigms, however, remain unclear but for the implicit reference to the notion of technological opportunities and their eventual exhaustion. The ultimate origin of technological change remains exogenous and a strong deterministic character is now added.

In this context, characterized by the decline of the heuristic power of the notion of trajectory and evolutionary frames, increasing attention is paid to the role of historic time. The evidence, provided by economic historians and historians of technology, makes clear the key role of technological cumulability and irreversibility, localized learning and local externalities (Dosi 1988; Freeman 1994).

More generally, it becomes evident that while evolutionary thinking provides a reliable and fertile approach to explain the selection process, it is less able to provide an explanation for the emergence of novelty and the regeneration of variation. Evolutionary thinking is not always able to disentangle from a framework where variations and occasional mutations

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are mainly the blind product of random processes determined by genetic recombination and drift, hence an appreciation of the role of the intentional components of decision-making, both in the generation of technological knowledge and in the introduction of innovations, seems necessary. The analysis of the role of the mutual interactions between the purpose-oriented and creative reaction of each agent and the changing conditions of the system, including the innovations being introduced by other agents, emerges as the approach by which technological change can be understood as an endogenous process (Antonelli 2003).

The analysis of the asymmetric effects of the introduction and diffusion of new technologies and of the structural determinants of the rate and the direction of technological change enables David (1990) to identify the path-dependent interplay between structural and technological change. Technological change is now viewed as a process able to alter the characteristics of the system and yet in itself the product of the characteristics of the system at each point in time. The application of the new general-purpose information and the communication technology in the United States does not materialize in terms of productivity, and is better understood in terms of system transition.

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5.5. Towards an economics of complexity

Complexity is emerging as a new unifying theory to understand endogenous change and transformation across a variety of disciplines, ranging from mathematics and physics to biology. Complexity favors the systemic approach in that the outcome of the behavior of each agent and of the system into which each one is embedded can only be understood as the result of the interaction between micro and macro dynamics. Complexity builds on a number of basic assumptions:

(1) *Heterogeneous agents.* Agents are characterized by distinctive and specific characteristics as well as being intrinsically heterogeneous.
(2) *Location matters.* Location in a multidimensional space, in terms of distance among agents and their density, matters and influences both behavior and performance.
(3) *Local knowledge.* Each agent has access only to local information and local knowledge, i.e. no agent knows what every other agent knows.
(4) *Local context of interaction.* Agents are localized within networks of relations, including transactions and feedbacks, which are specific subsets of the broader array of interactions that define their behavior.
(5) *Creativity.* Agents are creative, i.e. agents can follow some rules, but they can also change the rules. They do this in response to given feedbacks, according to their own specific characteristics and the features of local endowments, including the network of transactions and interactions into which they are embedded.
(6) *Systemic interdependence*. The outcome of the behavior of each agent is strictly dependent on the web of interactions that take place within the system. Hence at each point in time, the topology of the system, i.e. how the characteristics and structural interactions of the agents in their relevant multidimensional spaces are distributed, plays a key role.

Complex systems are characterized by non-ergodicity, phase transition and emergent properties. When non-ergodicity applies a little shock at a particular point in time, this affects the long-term dynamics of a system. Phase transitions consist in qualitative changes that can be determined by small changes in the parameters of the system. Emergent properties are properties of a system that apply at a specific level of aggregation of a system.

As Rosser (1999) and Foster (2005) show, the merging of the theory of complexity and economics contributes to the building of an economic theory of complexity based on non-ergodicity, phase transition and emerging properties. The integration of the rich and elaborated competence of economics in dealing with systemic analysis, although in a static context, can draw on complex system dynamics, especially when the role of historic time, and the intentional behavior of rent-seeking agents, are taken into account, and when an understanding of the economics of innovation is integrated.

According to the analysis of Durlauf (2005), the notion of path dependence is the specific form of complex dynamics applied to understanding economic systems as evolving systems that makes it possible to integrate into a single and coherent framework a number of relevant and complementary contributions.

Path dependence provides a unique and fertile analytical framework, which is able to explain and assess the ever-changing outcomes of the combination of and interplay between factors of continuity and discontinuity, growth and development, hysteresis and creativity, routines and 'free will', which all characterize economic action in a dynamic perspective that is also able to appreciate the role of historic time.

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According to the analysis articulated by Paul David in a long sequence of contributions dating back to the 1975 book, path dependence is an attribute of a special class of dynamic processes. A process is path-dependent when it is non-ergodic and subject to multiple attractors: 'Systems possessing this property cannot shake off the effects of past events, and do not have a limiting, invariant probability distribution that is continuous over the entire state space' (David 1992, 1; 1988, 1994, 2007).

Indeed, historic analysis and much empirical evidence in economic growth and specifically in the economics of innovation and new technologies confirm that these characteristics apply and are most relevant to understanding the laws of change and growth of complex systems. Path dependence is the specific form of complex system dynamics most apt to understand the process and the outcomes of the interactions among myopic agents embedded in their own context and constrained by their past decision, yet endowed with creativity and able to generate new knowledge by means of both learning and intentional innovative strategies, as well as through structural changes.

The notion of ergodicity deserves careful examination. When a process is non-ergodic, initial conditions (and events that occur at early points along the path) typically exert strong effects on its development and on the final outcome. Past dependence, or 'historicity,' is an extreme form of non-ergodicity. Historic, as well as social and technological, determinism fully belongs to past dependence. Here, the characteristics of the processes that are analyzed and their results are fully determined and contained in their initial condition. In the theoretical economics of innovation, this extreme (some would say degenerate) form of path dependence has often been assumed: the epidemic models of the diffusion of innovations and the notion of innovations 'locked in' a technological trajectory are typical examples of the deterministic representation of essentially stochastic technological and social phenomena. As such, these non-ergodic models are analytically informative but empirically uninteresting. The process takes place within a single corridor, defined at the outset, and external attractors cannot divert its route, nor can the dynamics of the process be altered by transient random disturbances in its internal operations.

Path dependence differs from deterministic past dependence in that irreversibility arises from events along the path, and it is not only the initial conditions that play a role in selecting from among the multiplicity of possible outcomes. The analysis of a path-dependent stochastic system is based on the concepts of transient or 'permanent micro-level' irreversibilities, creativity and positive feedback. The latter self-reinforcing processes work

both through the price system and by means of non-pecuniary externalities shaped by social, non-market interactions. The conceptualization of stochastic path-dependence can be considered to occupy the border region between a view of the world in which history is relevant only to establish the initial conditions, but has no bearing on the developments of the process and another where the dynamics unfold deterministically.

Path dependence is the conceptualization of historical dynamics in which one 'accident' follows another relentlessly and unpredictably and yet the past narrows the scope of possible outcomes, shaping the corridor into which the dynamics takes place. Path dependence gives economists the scope to include historical forces without succumbing to naive historical determinism. At the same time, it makes it possible to reduce the ray of the relevant spaces into which the system is likely to move at each point in time. The understanding of the historic forces that shape the dynamics of both individual agents and aggregate systems provides a clue to foresee, with some degree of indeterminacy, the future developments of a dynamic process. In so doing, path dependences make it possible to substitute the deterministic fallacy of general equilibrium analysis with the stochastic understanding of long-term dynamic processes (Rosenberg 1994, 47–61).

An important distinction emerges here between path-dependent innovation and path-dependent diffusion. The former takes place when the path along which the firm acts is determined by the irreversibility of its production factors and by the accumulation of competence and tacit knowledge based on learning by doing and learning by using. In this case, the switching costs firms face influence the choice of the new technology when, because of changes in relative factor costs or in the levels of output, they try to change the levels of their inputs. The latter applies to the choice of the new technology as it is shaped by market conditions. Interdependence among users leads to increasing returns in adoption so that technologies that have been adopted by a large share of prospective users have a greater chance of winning out in the selection process and spreading throughout the rest of the system. The notion of path dependence elaborated by David (1975, 1988, 1992) belongs to the first case: firms are induced to follow a path of technological change by their internal characteristics. The notion of path-dependence adoption elaborated by Arthur (1989) and David (1985) applies to the choice of one new technology among many possible ones and clearly belongs to the second case: new technologies are sorted out by increasing returns to adoption at the system level. The distinction between internal and external path dependence is also crucial. In the first case, the emphasis is on the role of factors internal to each firm in shaping their path of innovation and change. In the latter, much more attention is instead given to the role of external factors, including feedbacks (Arthur 1999).

According to North (1997), the introduction of innovations takes place as the result of disequilibrium conditions of the system and reproduces new disequilibrium conditions. Technological change is now the endogenous outcome of a disequilibrium condition that has little chance to converge toward a new equilibrium. In fact, equilibrium and technological change emerge as opposite extremes: equilibrium is possible when no technological change takes place, and vice-versa.

Hence Mokyr (1990) can articulate the view that technological change is a form of systemic, dynamic, stochastic and finite increasing returns that lead to punctuated growth. Technological change in fact takes place when a number of highly qualified, necessary conditions apply. As Metcalfe, Foster, and Ramlogan (2006) document empirically the successful introduction of technological change is the fragile result of a complex set of necessary and complementary conditions where firms adapt continually to the changing conditions of their environment.

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A strong common thread links the analyses developed with the notion of life cycle and technological trajectory and the notion of path dependence. Only the latter, however, provides a theory to understand why and how technological change takes place sequentially along axes defined in terms of complementarity and cumulability, both internal and external to each firm. From this viewpoint, the technological path represents significant progress with respect to both the technological trajectory and the life cycle.

Path dependence applies to each agent and at the system level: hence, we can identify and articulate an individual and a systemic path-dependence.

Individual path-dependence provides the tools to understand the combination of hysteretic, past-dependent factors, such as the quasi-irreversibility of tangible and intangible production factors, stock of knowledge and competence, and localized learning, with the generative relationships and creative reactions that make possible, at each point in time, a change in the direction of the action of each agent, including the introduction of innovations. At the firm level, the generation of knowledge shares the typical characteristics of a path-dependent process where the effects of the past, in terms of accumulation of competence, mainly based on processes of learning in a localized context and interaction with a given structure of agents, exert an influence and yet are balanced by the specific creativity that is induced by the changing conditions of the system.

Antonelli (1997, 2007) provides a systematic account of much recent advances in the economics of innovation toward the economics of complexity. Firms innovate when faced with changes in the expected state of the world as generated by changes in both product and factor markets. Innovation is induced by the mismatch between unexpected events that myopic agents cannot fully anticipate and the irreversible decisions that need to be taken at any point in time. Firms induced to innovate by irreversibility and disequilibrium in both products and factor markets search locally for new technologies. The direction of technological change is influenced by the search for new technologies that are complementary to existing ones. This is all the more plausible when the introduction of technological changes is made possible by the accumulation of competence and localized knowledge within the firm. In this context, the introduction of innovations and new technologies is the result of a local search, constrained by the limitations of firms to explore a wide range of technological options. Procedural rationality pushes firms to limit the search for new technologies in the proximity of techniques already in use, upon which learning by doing and learning by using have increased the stock of competence and tacit knowledge. The rate of technological change in turn is influenced by the relative efficiency of the search for new technologies. This dynamics leads firms to remain in a region of techniques that are close to the original one and continue to improve the technology in use.

Firms are better able to change their technologies when, as a result of effective communication systems, local externalities can turn into collective knowledge; when high levels of investments can help the introduction of new technologies; when an appropriate institutional system of interaction between the academic community, public research centers and the business community is in place; when industrial dynamics in product and input markets can induce localized technological changes that in turn affect the competitive conditions of firms; when stochastic processes help the creative interaction of complementary new localized kinds of knowledge and new localized technologies to form new effective technological systems; when the dynamics of positive feedback can actually implement the sequences of learning along technological paths, as well as the interactions between innovation and diffusion. Such a set of dynamic and systemic conditions has strong stochastic features and is available in finite conditions: the process is unlikely to go on indefinitely until all possible combinations have been exhausted.

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The architectures of the system into which firms are localized exert a key role in shaping the dynamics both at the aggregate and the individual level. The structure of interactions, the networks of cooperation and communication, the flows of technological externalities, the structure of the markets for products and processes and the forms of competition that prevail in each of them, the geographical distribution of firms, their density in regional and technological spaces, the forms of organization within and among firms, the institutional context are the meso-economic carriers of history and, as such, embody the memory of the system. They change through time, albeit at a slow rate, as a result of the dynamics of agents and of the aggregate. The meso-economic characteristics of the system act as a filter between the dynamics at the individual and the aggregate levels (Dopfer 2005).

According to Krugman (1994) rugged landscapes in geographical, technological, knowledge, market and product space are at the same time the consequence and the determinants of complex dynamics. Path-dependent complexity makes it possible to pay attention to the structural characteristics of the system in terms of the distribution of agents in the different space dimensions, and to appreciate the architecture of the relations of communication, interaction and competition that take place among agents in assessing the rate and direction of technological change.

Some architectures are clearly more conducive than others. Architectures themselves are, however, the path-dependent products of intentional choices of location and mobility of agents, aware of the effects of their location in such a multidimensional space on their chances to generate and introduce timely, new and appropriate technological innovations. Systemic path-dependence explores the mix of past-dependent elements embedded in the structural characteristics of the system, such as endowments, industrial and economic structure, market forms and organization of the networks of communication and interaction in place, with changes to the architecture of the structure that collective action can introduce at each point in time.

The appreciation and identification of the structural conditions that shape economic systems and are conducive to the introduction and diffusion of new technologies is one of the main results of this line of analysis. The structure of social, non-market interactions is endogenous to the system itself: the architecture of knowledge networks is heavily influenced by the strategies of firms seeking to improve their location within systems of interactions. The exploration of the endogenous formation of coalitions within scientific communities provides useful insights both to increase the efficiency of scientific undertakings and to implement dedicated tools for science and innovation policy (David and Keely 2003; D'Ignazio and Giovannetti 2006).

In this context, the notion of the generative relationship introduced by Lane and Maxfield (1997) is crucial. Generative relationships are:

> constructive positive feedbacks – that – have an obvious counterpart: as the structure of agent/artifact space undergoes ripples of changes, new agents and artifacts come into being and old ones acquire new functionalities, so identities change – and hence, old interpretations of identity bear an increasingly strained relationship with observable actions, the facts of the world. Different agents respond differently: some respond to the resulting ambiguity by generating new attributions to make sense of experienced novelty, and so attributional heterogeneity increases – increasing further the possibility that participants in other relationships will achieve sufficient attributional diversity to become generative actually. (Lane and Maxfield 1997, 185)

Generative relationships lead to the introduction of innovations, and innovations feed structural change in agent/artifact space. The process takes place through a 'bootstrap' dynamics

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where new generative relationships induce attributional shifts that lead to actions that in turn generate possibilities for new generative relationships. The structural characteristics of the system in terms of the distribution of agents in multidimensional spaces, of their networks of communication, relationship and interactions qualified by aligned directedness, heterogeneity, mutual directedness, permissions and action opportunities, are key elements for the sustainability of the process.

The successful accumulation of new technological knowledge, the eventual introduction of new and more productive technologies and their fast diffusion are likely to take place in a self-propelling and spiralling process and at a faster pace within economic systems characterized by fast rates of growth where interaction, feedback and communication are swifter. In such special circumstances, the system can undergo a phase transition leading to the introduction of a new radical technological system.

The circular relationship between structure and innovation, and the conduct and performance of firms, are indeed influenced by the structure of the system as it stands at time t, but in turn they exert strong influences on the characteristics of the structure at time $t + 1$, with the introduction of innovations. A new structure is determined and in order to readjust to it, firms elaborate new strategies that include the introduction of further innovations. The understanding of this recursive relationship paves the way to grasping the basic elements of the continual and dynamic system of feedback between the conduct and performance of firms, the rate and direction of technological change and structural change with a growing awareness of its evolving and historic characteristics.

As Arrow (2000) suggests, in these circumstances, the generation of new technological knowledge and the introduction of new technologies can be viewed as the cause and the consequence of punctuated economic growth and dynamic increasing returns.

Innovation is both the result and cause of out-of-equilibrium conditions. A clear continuity, ever since biological grafts came into the trajectory and finally the systemic network approach, confirms that innovation can only be understood in an analytical context that accepts the integration of the analysis of firms and agents that are continually pushed away from potential equilibrium conditions and attempts to react to the unexpected conditions of both product and factor markets by means of the introduction of new products, new processes, new organizational modes and new markets.

6. Conclusion

Economics of innovation is a distinctive area of specialization within economics, with a well-defined set of competences about the origins, causes, characteristics and consequences of the introduction of technological and organization changes in the economic system. At the same time, however, economics of innovation pretends to be one of the main pillars of the emerging economics of complexity.

It is the result of a long process. The starting point is indeed the discovery of the large portion of unexplained economic growth that only technological change can be credited for. The attempt to provide an economic explanation for technological change able to integrate analysis of the effects and causes of the introduction of innovations has led to the rediscovery of a number of forgotten dynamic paths provided by the history of economic thought. The four wide-ranging heuristic frameworks identified are: the classical legacies of Adam Smith and Karl Marx, the Schumpeterian legacy, the Arrovian legacy, and the Marshallian legacy eventually implemented by evolutionary approaches leading to complexity.

As soon as assumptions about the exogeneity of production (and utility) functions are relaxed and agents are considered both intelligent and endowed with a specific form of

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creativity that makes it possible to endogenously change the basic features of the utility and production functions and hence tastes, preferences, technologies and routines, the relevance of general equilibrium analysis declines. It is difficult to conceive a system of future prices that is able to take into account the introduction of all possible new technologies in a given time horizon. In fact, there is no longer a single attractor as firms are now credited with the capability of generating their own technological knowledge and changing their technologies, and not only varying either the quantity they produce or the prices they charge.

Identification of the four lines of investigation is the result of the progressive matching between a specific legacy of the history of economic thought and a specific area of investigation considered in the history of economic analysis. The classical legacy provided the basic inputs to elaborate a theory of economic growth based on the intentional introduction of technological innovations. The Schumpeterian legacy proved fertile to analyze the role of innovation within oligopolistic rivalry and made it possible to appreciate the role of entrepreneurship as a basic engine for the continual introduction of new technologies. The Arrovian legacy provided the first elements eventually enriched in a fully fledged analysis of the economic characteristics of knowledge from an economic viewpoint. Finally, the Marshallian legacy has led to the emergence of an evolutionary approach, eventually articulated in new complexity theory, that makes it possible to understand the process of specialization and structural change, based on the interplay between heterogeneity, complementarity and competition that characterizes the innovation process.

Each of the four approaches has a clear focus and a distinctive area of investigation. In the second part of the twentieth century they evolved in parallel with a process of specialization and consolidation of their respective areas of expertise. In a second step, however, an increasing number of lateral and horizontal contributions have been made. As a consequence, a quite consistent body of knowledge articulated in a portfolio of analytical tools has emerged out of the convergence of the four approaches with the progressive integration of the different fields of investigation.

This seems to be the context in which the analysis of the conditions of dynamic efficiency can be considered so that it can become one of the key aims and scopes of contemporary work in economic theory. The merging of complex dynamic theory with a theory of the agent based on subjective optimization implemented by the necessary consideration for creative choices in a context characterized by intrinsic heterogeneity of firms, can be productive both for economics and for building a more articulated theory of complex system dynamics.

Economic systems are more and more considered as complex dynamic mechanisms that are able to grow and have differentiated levels of dynamic efficiency. In turn, such levels of efficiency are the outcome of the behavior of heterogeneous agents and of the structure of their relations, in that they have a differential capability to change the rules and the network of interactions. Hence they are able to generate new technological knowledge and to introduce new technologies.

The notion of path dependence provides one of the most articulated and comprehensive frameworks from which to move toward an analysis of the conditions that make it possible to conceive the working of an economic system where agents are able to generate new technological knowledge, introduce new technological innovations and exploit endogenous growth. The notion of path dependence can be considered the analytical form of complexity most apt to understand the dynamics of economic systems where heterogeneous agents are characterized by some level of past dependence, as well as by local creativity, interdependence and limited mobility in a structured space that affects their behavior but is not the single determinant.

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Path dependence is an essential conceptual framework that goes beyond analysis of static efficiency and enters the analysis of the conditions for dynamic efficiency. It applies to each agent, in terms of the quasi-irreversibility of their own endowment of tangible and intangible assets, networks of relations in both product and factor markets, stock of knowledge and competence, and to the system level in terms of general endowments of production factors, industrial and economic structure, and the architecture of the networks in place.

The identification and articulation of individual and system path-dependence makes it possible to catch the basic laws of the continual interaction between the hysteretic effects of past dependence, both at the agent and at the system level, and the feedback dynamics that allows the intentional conduct of the creative agent to change both the course of their actions and the characteristics of the structured space. In so doing, path dependence retains the positive contributions of complex dynamic system methodology, and at the same time has the capability to overcome the intrinsic limitations stemming from its origins built on natural sciences where human decision-making is not considered. Indeed, the notion of path dependence is one of the main forays in the challenging attempt to apply the emerging theory of complexity to economics.

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